The Anatomy of the Principal Effect

Teachers, classrooms, and families under mandatory turnover

Paper idea for discussion. October 2026

What does a school leader change, and how far does it reach?

Principal value added on test scores is well documented. The process behind it, how leadership affects teachers, classrooms, and families, is not.

What we know

Magnitude	One SD of principal quality: 0.05 to 0.10 SD of achievement (Branch, Hanushek, and Rivkin, 2012; Böhlmark, Grönqvist, and Vlachos, 2016); 0.03 in math and 0.01 in reading once time-varying school shocks are removed (Bartanen, Husain, and Liebowitz, 2024).
Turnover	A change of principal lowers achievement by 0.03 SD and raises teacher turnover (Bartanen, Grissom, and Rogers, 2019).
Mechanism	Personnel: effective principals retain high-value-added teachers and dismiss low-rated ones (Grissom and Bartanen, 2019; Muñoz and Prem, 2024); they change the composition of the staff (Bartanen and Grissom, 2023).
Open	Effects on the same teachers' practice and wellbeing, on classrooms, and on families, under exogenous turnover.

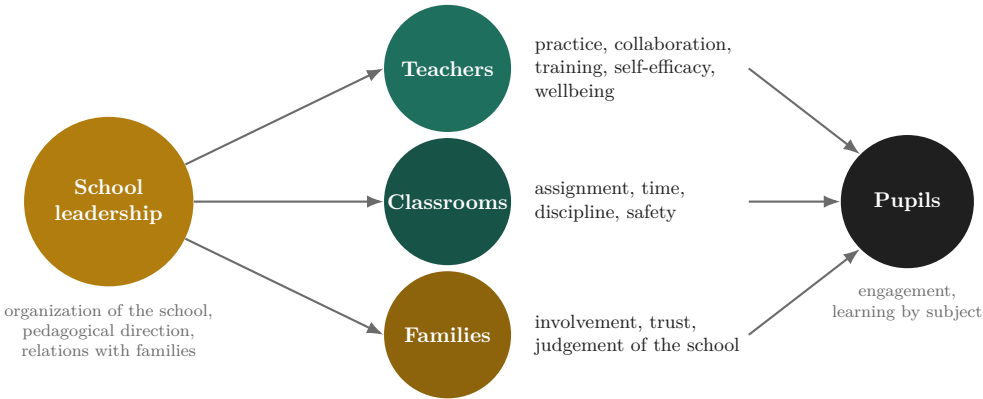
Recent work

Outcomes

Teachers	Classroom practice, collaboration, training, self-efficacy, wellbeing.
Classrooms	Allocation of teachers and support hours across groups, discipline, safety.
Families	Involvement, satisfaction with teachers and school, perceived quality.
Pupils	Engagement, perceptions of the school, standardized achievement by subject.

All four are observed for every school and year. The first three are outcomes in their own right and the channels to the fourth.

Causal map



Exogenous variation enters at the leadership node.

Setting: Madrid public primary schools

Principals	Teachers selected from the staff for four-year terms. They do not hire, dismiss, set pay, or set the curriculum.
Implication	A change of principal changes leadership and organization on a fixed staff. The personnel channel is shut by design.
Mandatory exits	Compulsory retirement at 65. At most two renewals: twelve years in office. Both dates are known years in advance.
Successors	Come from the same staff. Observed as classroom teachers in the year before appointment.

Identification: mandatory exits



Treatment	A principal exit forced by the age or tenure rule in 2016, 2017, or 2018.
Control	Schools whose principal has not yet reached either limit.
Assumption	The date of a mandatory exit is unrelated to contemporaneous school shocks. Testable on pre-trends.
Excluded	Voluntary exits (resignation, non-renewal). Estimated separately as the selection benchmark.

Event study

τ_s: year of the mandatory exit in school s. i: a teacher, classroom, pupil, or parent.

$$Y_{ist} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}[\, t - \tau_s = k \,] + \varepsilon_{ist}$$

β_k, $k \geq 0$	Effect k years after the exit, relative to the year before.
β_k, $k < 0$	Pre-trend test: zero if the exit date is exogenous.
Inference	Not-yet-treated schools as controls; staggered-robust estimator; clustered by school.

Effect of a change of principal

$Z_{st} = \mathbf{1}[t \geq \tau_s]$: the rule binds. $D_{st} = \mathbf{1}[\text{new principal}]$: the realized change.

$$D_{st} = \pi Z_{st} + \alpha_s + \lambda_t + u_{st}$$
$$Y_{ist} = \beta \widehat{D}_{st} + \alpha_s + \lambda_t + \varepsilon_{ist}$$

β	Effect of a new principal where the rule forced the change.
π	Equal to one for retirement. Below one for the tenure limit: the incumbent may be re-selected.
Within unit	Teacher fixed effects for the same teacher under two principals. Lagged score for the same pupil in grades 3 and 6.

Variance of principal effects

$Y_{st} = \mu_s + \theta_{p(s,t)} + e_{st}$, where θ is the effect of the principal in office.

$$\mathrm{Var}\big(\Delta Y_{st} \mid Z_{st} = 1\big) - \mathrm{Var}\big(\Delta Y_{st} \mid Z_{st} = 0\big) = 2\,\sigma_\theta^2$$

σ_θ^2	Dispersion of principal effects on each outcome, without ranking principals (Jones and Olken, 2005).
Assumption	e_{st} independent of Z_{st}. Same pre-trend test.
Use	Separates "principals do not matter" from "effects cancel on average".

Heterogeneity by pre-determined characteristics

$$Y_{ist} = \alpha_s + \lambda_t + \beta\, D_{st} + \delta\, D_{st} \times W_s + \varepsilon_{ist}$$

W_s	Measured before appointment: the successor's practices and wellbeing as a teacher in $\tau_s - 1$; the selection committee's score of their project; the staff's rating of the outgoing principal in $\tau_s - 1$.
δ	Variation of the effect with who leaves and who arrives.
Check	W_s uncorrelated with pre-trends.

Outcomes and expected timing

Domain	Outcome	Source	When
Classrooms	Assignment of teachers and support hours across groups	Timetables, class composition	Year 1
Classrooms	Discipline and climate	Pupil and teacher reports	Years 1–2
Families	Involvement, trust, judgement of the school	Parent survey	Years 1–2
Teachers	Self-efficacy, wellbeing, collaboration, training	Teacher survey	Years 1–2
Teachers	Classroom practice, within teacher	Teacher survey	Years 2–3
Pupils	Engagement and achievement by subject	Pupil survey, tests	Last

Pre-registered indices by domain.

Data

In hand	Universe of Madrid public primary schools, 2015 to 2018. Principal questionnaire. Teacher questionnaire: practices, wellbeing, training. Timetables. All pupils by classroom: scores by subject, perceptions. All parents by classroom: involvement, satisfaction.
To collect	Principal appointments, renewals, and exits with cause (BOCM). Committee score of each project. Teacher assignment lists. School plans from archived websites. Public school-level results from 2022.

Open questions

- **Which outcomes first.** Which domains should respond in the first year of a new principal.
- **Data requirements.** Stable teacher identifier across years; principal's age and years in office; survey items rating the principal.
- **Sample.** Number of mandatory exits in 2015 to 2018; access to appointment records and later years.

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References III

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Appendix: recent work on school and public-sector leaders

Title	Journal	Year
Principal effects on teacher working conditions (Rahaman and Koedel)	CALDER WP	2026
Rethinking principal effects on student outcomes	J. Public Economics	2024
Managers' productivity and recruitment in the public sector: the case of school principals	AEJ: Economic Policy	2024
Personnel management and school productivity: evidence from India	Economic Journal	2024
The lasting impacts of middle school principals	NBER WP	2024
Managers in public schools (Tsao)	Princeton JMP	2024
Effects and mechanisms of CEOs quality in public education	Economic Journal	2023
School principal race, teacher racial diversity, and student achievement	J. Human Resources	2023
Managers and productivity in the public sector	Econometrica	2022

Tsao: event studies of principal moves and retirements, Kentucky and Texas; effects on scores, absenteeism, teacher retention. Rahaman and Koedel: variance of principal effects on 20 teacher-reported working conditions, Illinois.

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